



CS2006 TASK 2 COURSEWORK

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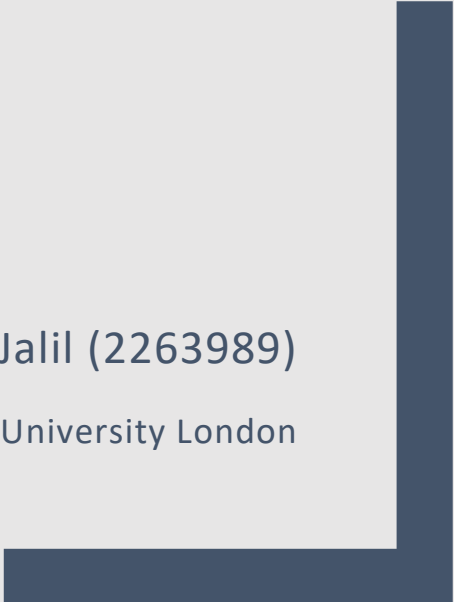


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INTRODUCTION

As required by Module CS2006 Task 2, the accompanying report provides a critical explanation of the Simulation Model and its constituent parts. My thought process, model building, model refining, trial calculations, data gathering, outcome analysis, resource management, reflective writing on learning and development, and other steps are all covered in the report.

The task provides a business scenario of a company named “BIPC” that works with three suppliers A, B, and C. We are developing a simulation model for the company, detailing the transactions and interactions amongst its various parts, the data flow and simulation within the model, and the conditions we apply to meet task 2's requirements. We use a predetermined set of data to develop our model and fill it in as needed. Every student has a different data set. My particular set is listed below.

Student Information			Order Arrivals				Product type probabilities		
First Name	Family Name	ID	BIPC	Supplier A	Supplier B	Supplier C	SHOES	BAGS	CLOTHES
Qasim	JALIL	2263989	10	20	15	10	10	35	55

BIPC LTD

Order arrival – **exponential** – see Yellow column for your data

OrderChecking work centre – average (5)

OrderHandling work centre – average (10)

OrderUpdates work centre – average (10)

Supplier A

Order arrival – **exponential** – see Orange column for your data

OrderType work centre – average (15)

P_OrderProcessing work centre – average (20)

C_OrderProcessing work centre – average (25)

Supplier B

Order arrival – **exponential** – see Green column for your data

OrderType work centre – average (15)

P_OrderProcessing work centre – average (10)

C_OrderProcessing work centre – average (15)

Supplier C

Order arrival – **exponential** – see Red column for your data

OrderType work centre – average (20)

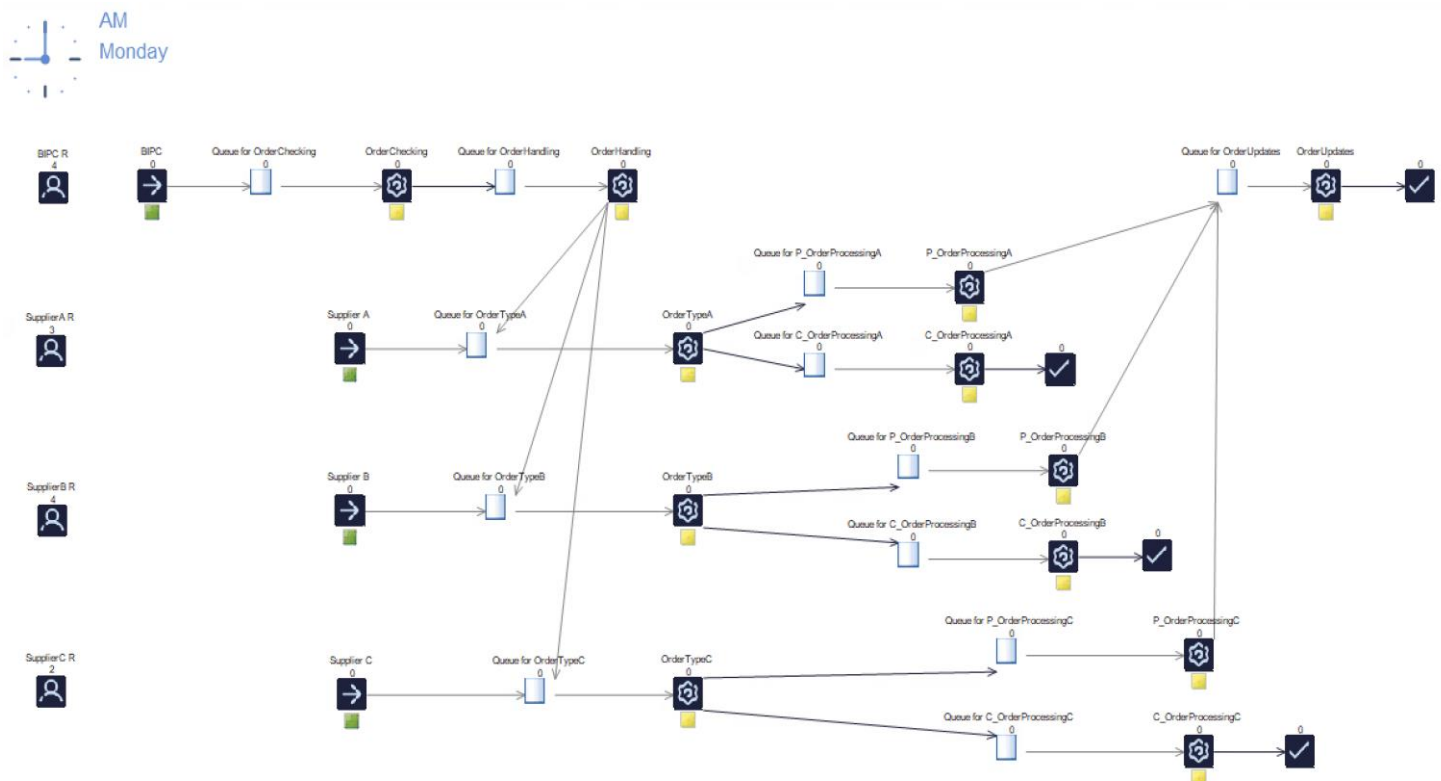
P_OrderProcessing work centre – average (10)

C_OrderProcessing work centre – average (15)

The data set states the individual values (for example: Exponential 10 at BIPC Order Arrival) I need to assign to my Work Centers to get accurate results.

MODEL DEVELOPMENT

One of the most crucial parts of this coursework is developing your model since the accuracy of your model will determine the Results and how the data is gathered. After generating three iterations and honing them to produce the final model, I developed the simulation model. Included in the model are the Suppliers "A", "B", and "C" along with their elements and goods, as well as the parent company "BIPC." Below is a picture of the model:



This is the final generated model. The model consists of the companies as mentioned above and its working is explained below.

WORKING:

The model starts off with the “BIPC” parent company’s Order Arrival gate where the company receives orders. The orders are received and stored and are sent to the “OrderChecking” Work center where the orders are checked. The orders are then sent to the “OrderHandling” Work center which plays a crucial part in the model. This Work Center is responsible for distributing the orders to the specific Suppliers. The orders consist of “Shoes”, “Bags”, and “Clothes”.

Given that the problem description makes this clear, we can infer that:

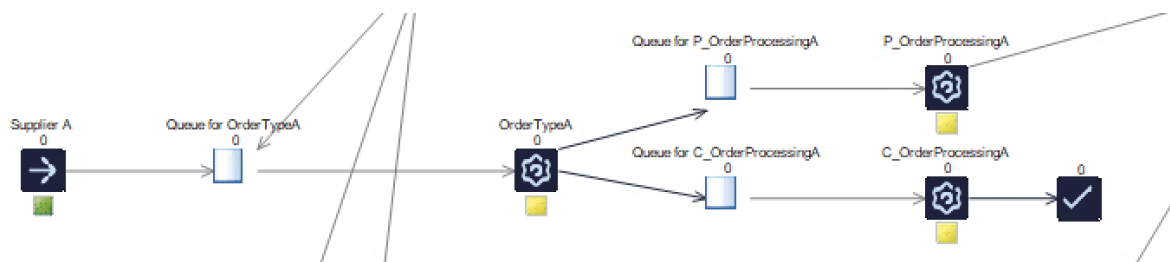
- “Shoes” are distributed to Supplier A.
- “Bags” are distributed to Supplier B.
- “Clothes” are distributed to Supplier C.

*(The details of how they are distributed are mentioned later.)

Before moving further, it is stated in the scenario that the Suppliers also receive their orders from their customers which divides the Orders into two categories:

1. The Orders that were received from the Parent Company “BIPC”.
2. The Orders that were received by the Suppliers from their own Customers.

SUPPLIER A:



The working goes as follows: Supplier A receives orders from the customer and the BIPC and they are stored in the Queue for OrderType A. Both orders are then moved to the Work Center where a special condition is applied, this condition distributes the Parent order (BIPC order) to the Order processing work Center for the parent company (P_OrderProcessingA), and the Customer Order is sent to the Work Center (C_OrderProcessingA). This is done by the following conditions:

Destination	Condition					
Queue for P_OrderProcessingA	L_Products	...	=	1	...	+
Queue for C_OrderProcessingA	L_Products	...	=	0	...	+

The orders from the Parent company (BIPC) are processed and sent back to the parent company to the Work Center “OrderUpdates” where they are updated and the process terminates at the endpoint. The Customer Orders are processed and it terminates at the endpoint.

SUPPLIER B/C:

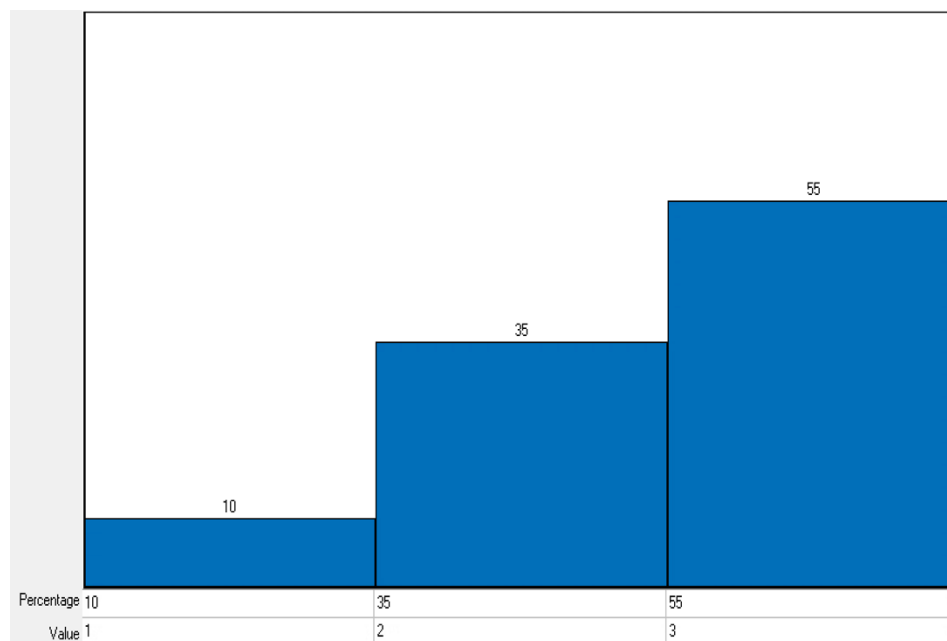
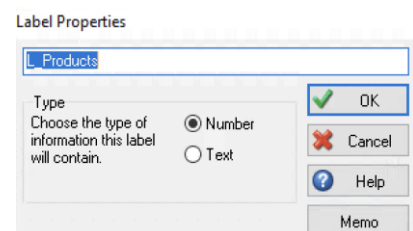
The process mentioned above for Supplier A is the same process that occurs for Supplier B and Supplier C. The difference only lies in the order received from the BIPC parent company as for supplier B it will be “Bags” and for Supplier C it will be “Clothes”. Rest is the same with the condition being targeted accordingly.

LABELS AND DISTRIBUTIONS:

The working of the Model is explained above however the working is not possible without the Label and Distribution profile. A label is used to attach the property of the order to a numeric value and populate the model. The Distribution Profile is created based on the probabilities provided by the Specific data provided individually as shown below.

Product type probabilities		
SHOES	BAGS	CLOTHES
10	35	55

- After the model is designed the first thing is to create a Label. In my case, the label is named “L_Products” with a numeric value. The label created is then assigned to the Order Arrival gate (starting point) of the BIPC company.
- The distribution profile is created next with the probabilities stated in the images above. My distribution profile is named “D_Products”. The distribution profile manages the flow of the items. The orders will be populated according to the percentages shown in the figure below.



This Distribution profile is then applied to the “OrderHandling” Work Center. This is one of the most important tasks as this will ensure that the orders are distributed accordingly. When the label and the distribution profile are attached to the

“Orderhandling” work center it will ensure and apply labels to the orders that are being distributed and works as follows:

- 1: Queue for OrderTypeA
- 2: Queue for OrderTypeB
- 3: Queue for OrderTypeC

- 1 numeric value is applied for orders going to Supplier A
- 2 number value is applied for orders going to Supplier B
- 3 number value is applied for orders going to Supplier B

This ensures that the correct orders are being distributed to correct work items thus enabling us to get the appropriate results we require.

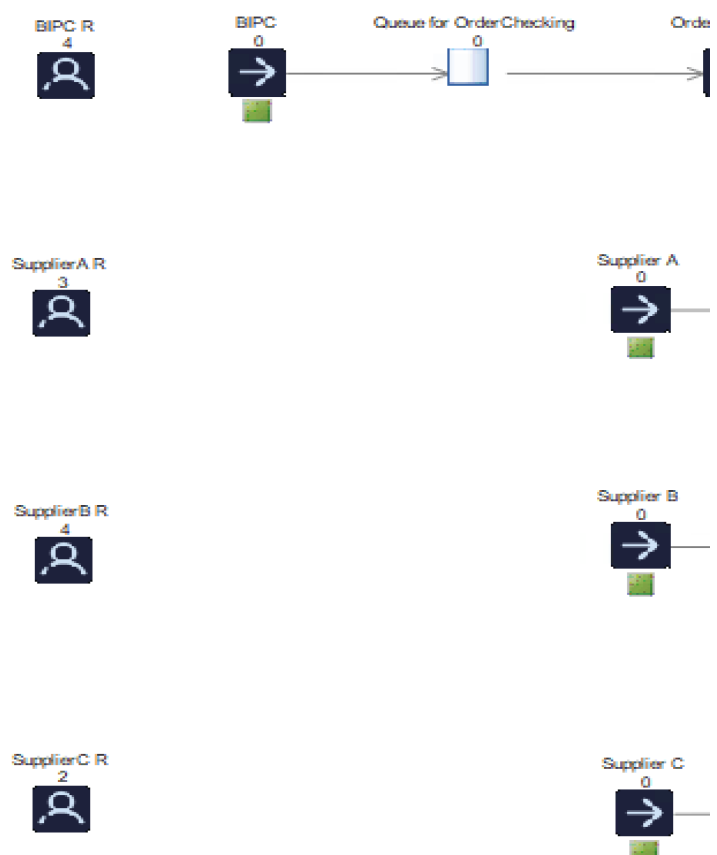
KPI:

KPI also known as ‘Key Performance Indicator’ is required for our result gathering and resource management. According to the problem statement, the required KPI for our Simulation model result is the Number of Work Completed at the Endpoint of the parent company which is BIPC. We acquire the KPI by right-clicking on the number of completed works and adding it to our results within the trial calculator. This KPI will be necessary further in the report when I will discuss the Result and table generation.

RESOURCES:

The knowledge I possess on Resources is not well rehearsed but upon my research and from my memory I believe resources are representations of humans working in a particular part of a system to simulate the workload. In the case of my simul8 model, these resources are humans that have been assigned to work items. BIPC and each part of the company for example Suppliers A, B, and C have their resources. These resources have been assigned to “Work Centers”.

BIPC R is the resource assigned to all the Work Centers in BIPC for example “OrderChecking”, “OrderHandling” etc. The same is the case for the resources of the Supplier branches. These resources play a key role as they are used to collect results from the generated Trial Runs. For each resource, the number of this type of resource can be changed. Thus, for our calculations, we simulate the



same scenario with a different number of “resources” to get a different “Number of Works Completed”.

On the other hand, we can also measure the resource utilization of how much load is on each resource to find the optimal work balance. For our Task 2 we are supposed to measure the resource utilization in the range of 70 to 75%.

EXPERIMENT RESULTS

After generating the model, and attaching labels, Distributions, and conditions for the optimal workflow, we populate the model and ready ourselves to take the results and later do analysis.

TABLE ONE:

STEP 1

All the resources are set to 1 and the simulation is run from the Trial Calculator. Inside the Trial Calculator, it is necessary to select the KPI we want to check to get the results.

The trial is run and after necessary runs, the results are displayed. We note down the “Number of Completed Works”.

STEP 2

We change the number of resources and set it to 2 for all of them (increment by 1). The Model runs again via the trial calculator and the changed value of “Number of Completed Works” is displayed. The number of work increases and it is noted down in the Table.

The “Number of Work Completed increases because the number of Resources have increased which means the more resources you have the more work can be carried out within the system.

STEP 3

Repeat the process by increasing the resources by 1 until the “Number of Works Completed” does not change. The point where the number does not change is noted. Make sure to note the number of Resources and “Number of Works Completed”.

The noted values in the table will give you the “MAXIMUM NUMBER OF STAFF NEEDED” for all the companies in the given KPI. I generated the chart for the same table to fulfill the chart requirement. Both the chart and Table are shown below.

Iteration	BIPC	Supplier A Resources	Supplier B Resources	Supplier C Resources	Work Completed KPI
1	1	1	1	1	38
2	2	2	2	2	99.5
3	3	3	3	3	113.33
4	4	4	4	4	113.33

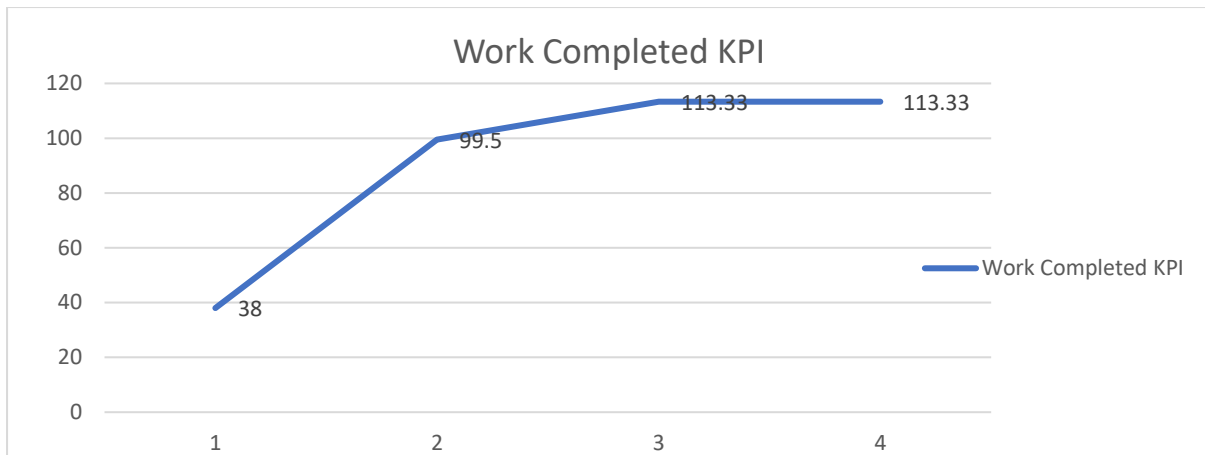


TABLE TWO:

The second required table provides us with the optimal number of Resources needed and resource utilization.

We start by taking the maximum number we got from table one. In my case, the number was 3 and for 3 as the number of resources for all the resources I got 113.33 as my “Number of Works Completed”. To check the optimal number we need to start reducing the number of resources from each part of the company starting with BIPC to see if the “Number of Works Completed” changes or not. If the number changes we stop and take the number that did not change the “Number of Works Completed” and move on to the next table. Keep repeating the process for each column marking the number that changes the KPI as **RED** and keeping the number before that as **GREEN**. Note down the values of the changed KPI and repeat the process until you reach the end of the last column.

This process will help you achieve the OPTIMAL NUMBER of resources needed.

Iteration	BIPC	Supplier A	Supplier B	Supplier C	KPI Work Completed
1	3	3	3	3	113.33
2	2	3	3	3	104.88
3	3	2	3	3	112
4	3	3	2	3	109.09
5	3	3	3	2	112.27
Optimal Number	3	3	3	3	113.33
Resource Utilisation%	63.74	68.59	62.05	55.11	
Adjustment 1	3	3	3	2	112.27
%	63.55	68.51	61.93	81.9	

RESOURCE UTILIZATION:

As you run the trials and select the option to see the resource utilization for each resource, the percentage of the resource utilization will be displayed after each trial run. Note down the percentage for the Trial Run with the Optimal Number of resources. In my result manager, I had approximately 63%, 68%, 62%, and 55% as the utilization for all 4 Resources I had for BIPC, Supplier A, B, and C respectively. This gives you an idea of what kind of workload is spread across the resources. In the next part, we will discuss how to manage the resources to get the percentage of utilization we desire in the range of 70 to 75%.

MANAGEMENT RECOMMENDATIONS

By examining Table 2, as previously said, we can conclude that when the ideal staff/resource ratio is used, resource utilization falls outside of the intended range which is 70 to 75%. To manage this range bracket, we will manage the resources in such a manner that benefits the range.

For this, we will analyze which part of the company lies the furthest away from the range. In my case, the resource utilization of “Supplier C” lies way below the range at 55.11%. Since we want more utilization out of the resources, we will decrease the number of resources at Supplier C. Running the trial will give us the adjusted utilization of 81.9% for Supplier C which is closer to the range.

Adjustment 1	3	3	3	2	112.27
%	63.55	68.51	61.93	81.9	

This adjustment will bring us 1 step closer to the desired range.

DISCUSSION:

In order to get the rest of the resource utilization in the desired range we either have to increase or decrease the resources accordingly. Let us run some trials to see the results we get by lowering each of the resources. Let us analyze whether they will fall into the range or not.

Since we need BIPC, Supplier A and B’s utilization to go up we will reduce resources for each by 1

	BIPC	Supplier A	Supplier B	Supplier C	KPI Work completed
Test Adjustments	2	2	2	2	99.50
%	89.03	96.73	87.51	81.9	

The table above gives us the reading of the utilization if we try to adjust the resources for BIPC, Suppliers A, and B. The readings above not only satisfy the range but also are quite higher than it. We adjust to make the readings closer to the range thus the “Test Adjustments” will not be considered and I will stick to the adjustment 1 for my table as it best suits the requirements.

REASON:

This occurs due to the simulation model not having a big data sample, since we are dealing with samples that are not that big thus it is hard to get the utilization accurately in the range. Each adjustment will have a significant impact on the resources and their utilization. Although the adjustment will decrease the “Total number of Completed Works in the given KPI” it is a small change that I am willing to overlook to make the utilization and work balance optimal for the resources. Thus, to conclude the management I will go for the adjusted number of staff with their utilization closest to the optimal range. Which is shown below:

Adjustment 1	3	3	3	2	112.27
%	63.55	68.51	61.93	81.9	

REFLECTIVE ESSAY

Question: **How important is it to use the correct (a) warm-up time, (b) result collection period, and (c) the number of trials in simulation experimentation?**

The factors mentioned above are key to getting accurate simulated results and ensuring certainty in your results. We will discuss each part in critical detail below while giving simulated examples:

WARM-UP TIME:

What is warm-up time?

Warm-up time refers to where the simulation model populates the model and stabilizes whilst running the simulation, the key point is the simulation runs without collecting any results or data that it is populated with.

The exact definition is stated in an article: “The warm-up period is the time that the simulation will run before starting to collect Results.” (McGregor, D.W. and Cain, M.J., 2004)

“This allows the queues and the machines to hopefully achieve a steady state (i.e., remove transient bias). If the simulation is set up with appropriate initial values, such as initial loadings and Work in progress, a warm-up period may not be required.” (McGregor, D.W. and Cain, M.J., 2004)

Some key points I found in an article regarding the importance of the correct warm-up period:

- Accurate: provides an accurate estimate of the length of the initial transient period.
- Reliable: consistently estimates the length of the initial transient.
- General: can be used across a range of output data types.
- Easy to implement (in Excel): does not require sophisticated statistical procedures.
- Requires minimum involvement from the user: on the basis that he/she does not have the necessary expertise.
- Varied: for instance, it was not seen as desirable to rely on purely graphical procedures, but a range of methods should be used.

Source: ([S. Robinson](#), 2005)

Based on my simul8 model if I change the warm-up period from 0 to 60 minutes as an example you can see the changes in the resource utilization and also the KPI work completed which clearly describes the accuracy of results depending on the correct warm-up periods as shown below:

End 1	Number Completed	109.14	114.67	120.19	
BIPC R	Utilization %	61.75	63.79	65.82	
SupplierA R	Utilization %	68.34	69.58	70.82	
SupplierB R	Utilization %	62.41	62.83	63.26	
SupplierC R	Utilization %	55.18	55.52	55.86	

RESULT COLLECTION PERIOD:

What is the result collection period?

It is known as the total time period in minutes/hours/days/weeks/months and years that the simulation will run for. It will collect the data over the period that has been set.

The exact definition is stated as “The Results collection period is how long the simulation will run before automatically stopping while collecting Results” (McGregor, D.W. and Cain, M.J., 2004)

The result collection period holds importance in the model as a lot of factors depend on this output.

- Trends and Variability in a system model depend on the correct result collection period as choosing a short-term collection period may miss important trends in a simulation trend
- Accuracy is a key feature that depends on the long-term collection period as the longer the simulation runs the more accurate the readings and results will be.

- Extended durations of data gathering yield greater quantities of data, but they also lengthen the simulation's runtime. It is imperative to strike a balance between computing resources and the duration of the collection period.

Coming back to my simul8 model it is clear that if I change the result collection period from 1 week to 1 month it will completely change the results of the KPI. This shows that the result collection period is necessary and should be accurate concerning the desired results. In task 2 we are required to keep the result collection to 1 week. Thus, the change in the result.

End 1	Number Completed	446.01	467.80	489.59	
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TRIALS RUN:

What is the number of trial runs?

The number of trial runs is a concept of running the simulation multiple times with different inputs and conditions to check the integrity of the simulated model. It can also be used to determine the results by running different random seeds.

The exact definition states: "A trial is a series of runs of the simulation. They are all performed with the same values for parameters but using different random number streams" (McGregor, D.W. and Cain, M.J., 2004)

The number of trial runs in simlu8 is used to reach the desired confidence limit in simul8. For Task 2, we are required to keep it as 5 as it gives us a 95% confidence limit.

OBJECTIVE:

A trial's objective is to verify the accuracy of the findings. You have mimicked a week in your organization after just one run. The result you got is from a single week ago. That could have been an average, terrible, or excellent week, just like in real life. Until we run a couple more, we have no way of knowing what kind of week it was.

A trial lasts for a few weeks, and the trial findings provide an overview of the outcomes throughout that time.

(This information is taken from Simul8 Website:

(https://www.simul8.com/support/help/doku.php?id=model_building_basics:trial)

Number of runs in trial:	5	Run Trial	Apply
Calculate Required Number of Runs		Extend	 Cancel

The Trials Calculator recommends a number of runs to use for trials.

The recommendation is based on your required precision of the confidence limits around the estimate of the mean for your simulation results (KPIs)

Required precision: % of the mean. Your selected (95%) confidence limits will each be within this % of the estimate of the mean.

CONCLUSION

In conclusion. The report provides a comprehensive overview of the simulation model development process, the experiment results, management recommendations, and a reflective essay. The simulation model for the company “BIPC” and its suppliers was meticulously developed and refined, with critical attention given to model components, data flow, and conditions. Experiment results were analyzed through the trial runs, leading to the key insights into resource utilization and optimal staffing levels. Management recommendations were provided to optimize resource utilization and achieve desired performance metrics. Reflective analysis highlighted the importance of factors such as warm-up time, result collection period, and the number of trials in simulation experimentation. Overall, the report serves as a valuable tool for understanding and optimizing business processes through simulation modeling.

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